Graph-KV on Colab: what the first runs show

Analysis of every run saved in the *graphkv_runs* folder on Google Drive, read on 29 October 2026. All runs used the paper's 8-billion-parameter model (*Tulu3-Block-FT*) on one NVIDIA L4 GPU.

Key findings in plain language

- **Our code does the same computation as the authors' code.** Replaying the authors' published answers word by word, our code would have picked the same word 99.2–99.8% of the time, and every difference is a rounding tie. Answering on its own, it gets the same questions right or wrong as the authors' code 95–99% of the time.
- **The paper's results came from a different passage layout than the one the paper describes** (the code's layout). This is now confirmed on two datasets.
- **On small samples, Graph-KV Full scored highest on 5 of 6 datasets**, but 39–50 questions per dataset are too few to rank methods (each accuracy is only known to about ±15 points). In the paper's own full results, plain sequential reading beats every Graph-KV variant on all four Table 3 datasets; Graph-KV's advantage is over Block-RAG.
- **Graph-KV was slower:** 23–46% more time per question than sequential reading in these runs.
- **In the three new experiments, Graph-KV's links between passages never helped** on the likelihood score: removing them made the score better in every comparison, and which passages were chosen as sources did not matter while all passages stayed visible.
- **Passages that read the sources keep almost none of the sources' facts.** With the sources removed, only 2 of 37 fact-chain questions whose answer was only in the sources were answered (0 of 37 without the links).
- **Caution:** the likelihood score used by the new experiments measures an answer format the model never uses, and it disagreed with real answers in the one place both were measured. The findings that rest on it are provisional (Section 9).

Background: the ideas you need for the tables

When a language model answers a question from several retrieved passages, the usual way is to paste the passages one after another (**sequential** reading). Each passage is then read in the light of the ones before it. This is accurate, but slow for long inputs, and the result depends on the order. A cheaper way encodes each passage on its own (**Block-RAG** or **parallel** encoding), so passages can be cached and reused, but then they cannot see each other.

Graph-KV (NeurIPS 2025, the paper being replicated) sits in between. It encodes the passages on their own, picks the most relevant few as **sources**, and lets every other passage (the **targets**) read the sources once. In graph terms, there is an edge from each source to each target. The question is then answered with all passages in view.

Terms used in this report

Term	Meaning
Passage / chunk	One piece of the text the model reads before the question (a retrieved document, or a window cut from the text).
Sequential	All passages pasted one after another; each one sees the ones before it. The standard way, and the most expensive.
Block-RAG	Each passage encoded on its own, laid out one after another. The paper's main cheap baseline.
Parallel	Each passage encoded on its own, all sharing the same positions. Our no-links control in E1 and E2.
Graph-KV Top-1 / Top-3	The 1 or 3 most relevant passages are sources; every other passage reads them once.
Graph-KV Full	Every passage reads every other passage once.
No-edge control	Same passages at the same positions as Graph-KV, but the targets never read the sources. Graph-KV minus this control is the effect of the links alone.

Term	Meaning
Sources removed (targets only)	After the targets have read the sources, the sources are thrown away and the question sees only the targets. Tests whether the targets absorbed what the sources said.
Accuracy	Share of questions whose correct answer appears in the model's reply (the paper's scorer).
Likelihood score	How surprised the model is by the correct answer (negative log-likelihood, in nats per word piece). **Lower is better.** Quick to compute (no text is generated), so E1–E3 mostly use it.
95% range	The range the true difference lies in with 95% confidence, from comparing the same questions under both conditions. If it does not include zero, the difference is real and not luck.
Rounding tie	The GPU stores numbers to about 3 significant digits (bfloat16). When two candidate words score within one or two rounding steps of each other, either can win. Not an error.
Published answers	The answers the paper's authors released for each method and dataset.

Datasets

Dataset	What the questions are like
2Wiki	Questions that need two Wikipedia facts. Four types: *comparison* (which of two, same country?), *bridge comparison* (compare through a linked person, e.g. whose director was born first), *compositional* (chain two facts to name something, e.g. the birthplace of a film's director) and *inference* (e.g. a grandparent).
HotpotQA	Two-fact Wikipedia questions (bridge and comparison).
NarrativeQA	Questions about books and films.
TriviaQA	Trivia questions.
MultiHop-RAG	Questions across several news articles: inference, comparison, temporal, and null (the answer is not in the articles).
MorehopQA	Multi-step questions needing 1 to 5 reasoning hops.

1. What was in the Drive folder

What we did

Each notebook cell saves its results to its own folder on Drive. Every run has a time limit so it cannot overspend; a run that hits its limit stops cleanly and continues from where it stopped when the cell is run again.

Table 1. Runs found, and whether they finished

Run (notebook cell)	What it measures	Questions	Status
E0: check against the authors (6a)	Does our code reproduce the authors' published answers?	30 + 30	Complete (2Wiki and HotpotQA)
Paper Table 3 sample (6b)	Accuracy on 4 datasets, 5 methods	39–40 of 50 per dataset	Stopped at its 100-minute limit
Paper Table 2 sample (6c)	Accuracy on 2 harder datasets, 5 methods	50 per dataset	Complete
E3, likelihood (7A)	Do targets absorb what the sources say?	300	Complete
E3, real answers (7B)	The same, by generating answers	126 of 150	Stopped at its 35-minute limit
E1 (8)	Four ways to choose the sources among 128-token windows	300 each	Complete
E2 (9)	Six ways to cut the text into passages	200 each	5 complete; 1 stopped at 138
E2 chunk-size sweep, Cora (9, optional)	Passage size; paper Table 4	–	Not run yet

Measured GPU time (without loading the model): Table 3 sample 99.5 min, Table 2 sample 95.3 min, E1 31 min, E2 33 min, E3 48 min. Total about 5.1 GPU hours, roughly 25 of the 200 monthly compute units.

What it tells us

- Everything planned up to experiment E2 ran. Three runs stopped at their time limits and can be resumed.
- About an eighth of the monthly compute budget has been used.

2. Does our code compute the same thing as the authors' code?

What we did

The authors published the answers their code generated. Those runs were *greedy*: at every step the model takes its single most likely next word. So if our code does the same computation, then when we feed in their answer word by word, their next word should also be our most likely word at every position.

We did this for 30 questions from 2Wiki (150 published answers, 5 methods) and 30 from HotpotQA (120 answers, 4 methods; the authors did not publish HotpotQA sequential answers). As a control, we also scored each answer under the *other* methods; those should agree less, which shows the check can tell methods apart.

Table 2. The authors' answers replayed through our code

Dataset	Authors' method	Words where we pick the same word (same method)	Answers matching at every word: ties allowed (strict)	Best match under a different method
2Wiki	Sequential	99.7%	30/30 (24)	96.7%
2Wiki	Block-RAG	99.6%	30/30 (22)	97.4%
2Wiki	Graph-KV Top-1	99.8%	30/30 (26)	98.9%
2Wiki	Graph-KV Top-3	99.7%	30/30 (23)	98.5%
2Wiki	Graph-KV Full	99.2%	30/30 (19)	97.9%
HotpotQA	Block-RAG	99.7%	30/30 (25)	96.3%
HotpotQA	Graph-KV Top-1	99.7%	30/30 (24)	97.3%
HotpotQA	Graph-KV Top-3	99.6%	30/30 (24)	97.8%
HotpotQA	Graph-KV Full	99.7%	30/30 (24)	97.8%

Table 3. Where the disagreements come from

	2Wiki	HotpotQA
Words checked	13,002	8,760
Words where we disagree	53	28
Largest score gap at a disagreement	0.5 (2 rounding steps)	0.25 (2 rounding steps)
Disagreements that are not rounding ties	0	0

Table 4. Which passage layout produced the authors' Graph-KV Top-3 answers?

Dataset	Layout in the authors' code (sources one after another)	Layout described in the paper (sources share positions)
2Wiki	99.7% of words; 30/30 answers	96.8% of words; 8/30 answers
HotpotQA	99.6% of words; 30/30 answers	97.0% of words; 10/30 answers

What it tells us

- Our code makes the same choices as the authors' code. Over 21,762 words, all 81 disagreements are rounding ties: the two best candidate words scored practically the same, and the authors' GPU broke the tie the other way.
- The check is sensitive. Scored under any other method, the same answers agree clearly less (by 0.9 to 3.4 points of words, and far fewer answers match at every word), so the matches are not a fluke.
- The paper says sources share one set of positions, but its published results were produced with the code's layout, which puts the sources one after another. Comparisons with the paper's numbers must use the code's layout, as all our replication runs do.
- The 2Wiki check, which ran out of GPU memory in the previous session, now completed after the fix.

3. The paper's Table 3 on a sample

What we did

The paper's Table 3 reports accuracy on four question-answering datasets for five methods. Running all ~35,000 questions would take hundreds of GPU hours, so each method answered a sample of about 40 questions per dataset, chosen to cover every question type.

Each cell gives three numbers: **ours**, the **authors'** published answers to the same questions (same scorer), and the **paper**'s printed number over all questions. The test of our code is ours against the authors'. The paper's number can differ from both simply because it covers different, many more questions. 50 questions per dataset were planned; the 100-minute limit stopped the run at 39–40.

Table 5. Accuracy (%): ours / authors' answers to the same questions / paper (all questions)

Method	NarrativeQA (40)	2Wiki (39)	TriviaQA (39)	HotpotQA (39)	Same right/wrong as authors
Sequential	80.0 / 77.5 / 63.2	79.5 / 79.5 / 77.3	69.2 / 69.2 / 76.6	79.5 / – / 77.4	117 of 118
Block-RAG	77.5 / 77.5 / 59.4	76.9 / 71.8 / 71.4	64.1 / 66.7 / 74.4	79.5 / 76.9 / 71.3	153 of 157
Graph-KV Top-1	75.0 / 75.0 / 62.0	74.4 / 74.4 / 73.6	66.7 / 66.7 / 75.2	**84.6 / 84.6 / 73.4**	155 of 157
Graph-KV Top-3	77.5 / – / 62.3	71.8 / 71.8 / 75.2	66.7 / 66.7 / 75.7	79.5 / 79.5 / 75.2	117 of 117
Graph-KV Full	**82.5 / 82.5 / 62.9**	**82.1 / 76.9 / 74.4**	**71.8 / 74.4 / 75.9**	82.1 / 82.1 / 76.2	154 of 157

– : the authors published no answers for this method (HotpotQA sequential: only 13 of 7,405; NarrativeQA Top-3). Bold: highest of ours on that dataset.

Table 6. Average over the four datasets

Method	Our sample	Paper (all questions)
Sequential	77.0	**73.6**
Block-RAG	74.5	69.1
Graph-KV Top-1	75.2	71.1
Graph-KV Top-3	73.9	72.1
Graph-KV Full	**79.6**	72.3

What it tells us

- Our answers are right or wrong on the same questions as the authors' in 696 of 706 cases (98.6%). The wording is identical in 73% of answers: a rounding tie early on changes the rest of the sentence, but rarely whether the answer is right.
- On this sample Graph-KV Full has the highest average, ahead of sequential reading. But with about 40 questions per dataset, each accuracy is uncertain by about ±16 points, so the sample cannot rank the methods.
- In the paper's own full results for this model, sequential reading beats every Graph-KV variant on all four datasets. Graph-KV's real gain is over Block-RAG (2–3 points): it recovers part of what is lost by encoding passages separately, without beating full sequential reading.
- Our NarrativeQA sample happens to be much easier than the full dataset (75–83% against 59–63% in the paper). The authors' own answers score just as high on it, so this is the sample, not an error.

4. The paper's Table 2 on a sample

What we did

The same procedure on the paper's two harder, multi-step datasets, with 50 questions each. This run finished.

Table 7. Accuracy (%): ours / authors' answers to the same questions / paper (all questions)

Method	MultiHop-RAG (50)	MorehopQA (50)	Same right/wrong as authors
Sequential	32.0 / 30.0 / 43.6	36.0 / 34.0 / 37.4	94 of 100
Block-RAG	40.0 / 38.0 / 43.3	40.0 / 44.0 / 37.9	93 of 100
Graph-KV Top-1	42.0 / 42.0 / 44.8	44.0 / 48.0 / 38.0	96 of 100
Graph-KV Top-3	40.0 / 44.0 / 45.8	46.0 / 44.0 / 38.1	95 of 100
Graph-KV Full	**48.0** / 48.0 / 46.4	**52.0** / 48.0 / 44.9	98 of 100

Table 8. Our accuracy (%) by question type (number of questions in brackets)

Method	MH-RAG inference (16)	MH-RAG comparison (17)	MH-RAG temporal (12)	MH-RAG null (5)	MoreHop 1 hop (20)	MoreHop 2 hops (19)	MoreHop 3 hops (7)	MoreHop 5 hops (4)
Sequential	62.5	5.9	16.7	60	55.0	26.3	28.6	0
Block-RAG	**87.5**	**11.8**	25.0	20	**70.0**	15.8	28.6	25
Graph-KV Top-1	75.0	0	33.3	100	60.0	21.1	57.1	50
Graph-KV Top-3	68.8	0	33.3	100	60.0	31.6	57.1	25
Graph-KV Full	81.2	0	**50.0**	100	60.0	**42.1**	57.1	50

What it tells us

- Our answers match the authors' on right/wrong in 476 of 500 cases (95.2%). This is a little lower than for Table 3 because these answers are longer (about 115–150 word pieces), which leaves more room for a rounding tie to send the answer elsewhere.
- Graph-KV Full scores highest on both datasets in the sample, with two caveats. On MorehopQA the paper checked Graph-KV Full's answers more leniently than the other methods'; we copied this to compare with the paper, so it flatters Full. Our earlier re-scoring of the authors' published answers showed that, scored the same way, Full's advantage in the paper shrinks from +6.5 points to between −3.0 and +1.0.
- On MultiHop-RAG, the paper's scorer was never released, so we used the benchmark's own scorer and the paper column is not comparable. Under that scorer, comparison questions score near zero for every method.
- Where Graph-KV helped in this sample: MultiHop-RAG's null questions (correctly saying the answer is not in the articles: 5 of 5 against 1–3 of 5) and MorehopQA questions with two or more hops. On one-hop questions Block-RAG did best. These groups are small.

5. How long each method takes

What we did

Every question in the Table 3 and Table 2 runs was timed on the L4 GPU, from reading the passages to the end of the answer.

Table 9. Seconds per question (and time relative to sequential reading)

Dataset	Sequential	Block-RAG	Graph-KV Top-1	Graph-KV Top-3	Graph-KV Full
NarrativeQA	8.5	8.1 (0.95×)	9.3 (1.09×)	9.4 (1.11×)	10.4 (1.22×)
2Wiki	5.7	6.4 (1.12×)	7.2 (1.26×)	7.1 (1.25×)	8.0 (1.40×)
TriviaQA	6.1	6.9 (1.13×)	7.3 (1.20×)	8.1 (1.33×)	9.0 (1.48×)
HotpotQA	5.3	5.9 (1.11×)	6.8 (1.28×)	7.4 (1.40×)	8.7 (1.64×)
MorehopQA	9.8	10.8 (1.10×)	11.5 (1.17×)	12.2 (1.24×)	12.9 (1.32×)
MultiHop-RAG	8.5	9.9 (1.16×)	11.8 (1.39×)	12.1 (1.42×)	14.7 (1.73×)
Average ratio	1.00×	1.10×	1.23×	1.29×	1.46×

What it tells us

- Graph-KV took 23–46% longer per question than sequential reading on average, and Full up to 73% longer.
- The extra step in which the targets read the sources costs more than Graph-KV saves at these context lengths. These runs also encode every passage afresh for each question; the time savings that block methods promise come mainly from reusing passage encodings across questions and from much longer inputs, which these runs did not test.

6. Experiment E3: do the targets absorb what the sources say?

What we did

Graph-KV's central idea is that when the targets read the sources, information flows into the targets. To test it, after the targets had read the sources we **removed the sources** and asked the question with only the targets in view. We compared this with the **no-edge control**: the same targets at the same positions, which never read the sources. If information flows, Graph-KV's targets should do better, above all when the answer was only in the sources.

Setup: 2Wiki questions; the passages are the 10 retrieved documents of each question; the retriever's top 3 are the sources. Measured two ways: the likelihood score on 300 questions (fast), and real generated answers on 126 questions (the run stopped at 126 of 150).

Table 10. Likelihood score, 300 questions (lower is better)

Condition	What the question sees	Word pieces kept	Score
Graph-KV Top-3	Sources and the targets that read them	985	8.98
No-edge control	Same passages and positions, no reading	985	8.68
Graph-KV, sources removed	Only the targets that read the sources	601	8.92
No-edge, sources removed	Only targets that never read the sources	601	**8.53**
Graph-KV, targets removed	Only the sources	422	9.14

Table 11. Comparisons on the same 300 questions

Question	Difference in score	95% range	Verdict
Do the links help? (all passages visible)	+0.30 (worse)	+0.24 to +0.37	Links hurt
Do the links help? (sources removed)	+0.40 (worse)	+0.31 to +0.49	Links hurt
What does removing the sources cost?	−0.06	−0.27 to +0.18	Nothing measurable
What does removing the targets cost?	+0.16 (worse)	+0.02 to +0.30	A small cost

Table 12. Real answers with the sources removed: accuracy (%)

Questions	Number	Graph-KV targets	No-edge targets	Difference (95% range)
All questions	126	42.9	41.3	+1.6 (−5.6 to +8.7)

Graph-KV alone got 10 questions right, the control alone 8.

Table 13. The same, for questions whose answer was only in the removed sources, by question type

Question type	Number	Graph-KV targets	No-edge targets
Comparison (which of two, yes/no)	15	100	87
Bridge comparison	19	84	84
Compositional (name a fact)	37	**5 (2 of 37)**	0 (0 of 37)
Inference	8	0	0
All four types	79	41.8	36.7

What it tells us

- Judged by real answers, the links made almost no difference: 42.9% against 41.3%, and the 95% range includes zero.
- The telling group is the fact-chain (compositional) questions whose answer was only in the removed sources. The targets that had read the sources answered 2 of 37. So the targets keep very little of the sources' specific facts.
- Comparison questions were answered well either way. Their answers (yes/no, one of two names) can often be worked out from the remaining passages, the model's own knowledge, or a lucky guess, so they say little about information flow.
- The likelihood score points the other way: it says the links make things worse (+0.40, a clear effect). Section 9 explains why we trust the real answers more.

7. Experiment E1: choosing the sources among plain text windows

What we did

In the paper the passages are whole retrieved documents, ranked by a retriever. Here we asked whether Graph-KV also works when the text is simply cut into equal windows of 128 word pieces, and how best to choose the top 3 windows that become the sources.

Four ways to choose: the dataset's **retriever** scores, **BM25** keyword matching, the model's own **attention** from the question to each window, and **random** choice as the control. 300 2Wiki questions, about 8 windows each, measured with the likelihood score. For each chooser we ran parallel encoding (no links), Graph-KV Top-3, and both with the sources removed.

Table 14. Likelihood score by way of choosing the sources (lower is better)

Chooser	Answer in the chosen 3	Parallel (no links)	Graph-KV Top-3	Graph-KV, sources removed	No-edge, sources removed	Effect of links, sources removed (95% range)
Retriever	60.3%	8.60	8.81	8.82	8.44	+0.38 (+0.29 to +0.48)
BM25	**66.7%**	8.60	8.81	8.72	**8.36**	+0.37 (+0.26 to +0.48)
Attention	57.7%	8.60	8.77	9.00	8.65	+0.34 (+0.24 to +0.45)
Random	44.3%	8.60	8.78	9.02	8.68	+0.34 (+0.23 to +0.45)

Table 15. Each chooser compared with random choice on the same questions (negative = better than random)

Condition	Retriever	BM25	Attention
Graph-KV Top-3, all visible	+0.03 (−0.03 to +0.09)	+0.03 (−0.03 to +0.09)	−0.02 (−0.07 to +0.03)
Graph-KV, sources removed	**−0.20** (−0.37 to −0.02)	**−0.29** (−0.45 to −0.13)	−0.02 (−0.18 to +0.14)
No-edge, sources removed	**−0.24** (−0.40 to −0.07)	**−0.32** (−0.46 to −0.18)	−0.02 (−0.17 to +0.12)

Parallel encoding does not depend on the chooser and gave identical scores in all four runs, as it should.

What it tells us

- BM25 keyword matching finds the window that holds the answer most often: 67% of questions, against 60% for the retriever, 58% for the model's attention and 44% for random choice.
- While all windows stay visible, it does not matter which windows become the sources: every chooser is within 0.03 of random. If the links carried useful information, choosing the right sources should have helped.
- Graph-KV Top-3 scored worse than parallel encoding, where the windows never read each other, with every chooser.
- Once the sources are removed, good choosers (BM25, retriever) help, but by the same amount with or without links. The gain comes from which windows remain, not from the graph.
- The model's own attention finds the answer window more often than random (58% against 44%), but scores no better.

8. Experiment E2: what should count as a passage?

What we did

Graph-KV treats each passage as a node of its graph. We cut the same text five ways: the original **documents**; **fixed** 128-piece windows; groups of **sentences** and of **paragraphs** of up to about 128 pieces; and **attention** cuts, placed where the model's attention across the boundary is weakest (about 128 pieces). BM25 chose the top 3. 200 2Wiki questions, likelihood score. A sixth way (attention cuts aligned to sentences) stopped at 138 questions.

Table 16. Likelihood score by way of cutting the text (lower is better)

Passages	Passages per question	Word pieces per passage	Answer in the top 3	Parallel (no links)	Graph-KV Top-3	Graph-KV, sources removed
Documents	10.0	94	65.5%	7.90	8.22	8.08
Fixed windows	7.8	119	70.0%	**7.87**	**8.05**	7.99
Sentences	5.1	**183**	80.0%	8.02	8.19	8.00
Paragraphs	4.5	**207**	83.5%	8.08	8.20	8.07
Attention cuts	7.4	128	69.0%	7.92	8.13	**7.87**

Table 17. Each way compared with whole documents on the same questions (negative = better than documents)

Condition	Fixed windows	Sentences	Paragraphs	Attention cuts
Parallel (no links)	−0.04 (−0.14 to +0.07)	+0.11 (+0.05 to +0.19)	+0.18 (+0.11 to +0.25)	+0.02 (−0.08 to +0.12)
Graph-KV Top-3	**−0.17** (−0.27 to −0.06)	−0.02 (−0.10 to +0.05)	−0.01 (−0.09 to +0.06)	−0.08 (−0.17 to +0.01)
Graph-KV, sources removed	−0.09 (−0.26 to +0.08)	−0.08 (−0.29 to +0.11)	−0.01 (−0.22 to +0.19)	**−0.21** (−0.43 to −0.02)

What it tells us

- The sentence and paragraph cutters overshot the target size (183 and 207 pieces instead of about 128), so their top 3 covered over half of the text. Their higher answer-in-top-3 rates come from bigger passages, not better boundaries. Only fixed windows, attention cuts and documents are fair comparisons.
- With everything visible, fixed windows suited Graph-KV best (0.17 better than documents, a real difference). With the sources removed, attention cuts did best (0.21 better, just clear of zero).
- Once more, Graph-KV with everything visible scored worse than parallel encoding with every way of cutting (by 0.13 to 0.31). With the sources removed, it scored about the same as parallel, although it keeps only 46–63% of the text.
- The unfinished attention-and-sentence run stayed within 0.14 of documents on its 138 questions.

9. How far to trust the likelihood score

What we did

To save GPU time, E1–E3 mostly measured how likely the model finds the correct answer (for example *Bratislava*) as the very first words of its reply. This takes one pass instead of generating a full answer. Four checks show how well it works here.

Table 18. Checks on the likelihood score

Check	What a good score would show	What we found
How surprised is the model?	Low for an answer the model would give	8–9 for the bare answer, against 0.05 for the model's own wording (measured when replaying the authors' answers, Section 2). The model opens with “Based on the…” (85 of 126 replies) or “I apologize, but…” (30), never with the bare answer.
Answer in the text vs nowhere in it	Much lower when the answer is in the text	Yes: about 7 against about 20. The score carries real signal.
Hide the passage holding the answer	Score gets worse	On 88 questions where BM25 hid the answer window and random choice kept it: no change (−0.10, range −0.43 to +0.24). In the reverse case (24 questions) the score improved by 1.7 to 2.1.
Agreement with real answers (E3)	Same direction	Opposite: the score says the links hurt (+0.40, clear); real answers say they help slightly (+1.6 points, not clear).

What it tells us

- The score knows when the answer is nowhere in the text, but it measures a reply format the model never uses, and it reacts inconsistently when the answer is hidden.
- Differences of 0.1–0.4 between conditions may reflect how each condition changes the model's phrasing rather than what it knows. The findings that rest only on this score are provisional: the links hurting (E1–E3), the chooser not mattering while everything is visible (E1), and the best way of cutting the text (E2).
- A better score would measure the answer inside the model's usual reply, for example after the words “Based on the provided search documents, the answer is”. Or the key comparisons can be checked with real answers (about 40 minutes per pair of conditions on 150 questions).

10. Recommended next steps

#	Step	Why	Cost
1	Fix the likelihood score (score the answer after a lead-in such as “Based on the provided search documents, the answer is”), then re-check the main E1 and E2 comparisons with real answers	The current score disagrees with real answers	~40 min per comparison
2	Resume the three runs that hit their time limits: Table 3 sample (10–11 more questions per dataset), E3 real answers (24 more), attention-and-sentence cuts (62 more)	Complete the planned samples	~45 min, with model loading (estimate)
3	Add a real-answer run of E3 with everything visible	Measures what removing the sources costs in accuracy	~40 min
4	Run what has not run yet: the passage-size sweep (32, 64, 256 pieces) and the Cora classification (paper Table 4)	Planned experiments	~45 min; ~1 h

Source data: Google Drive, MyDrive/graphkv_runs (run summaries, per-question records and run settings). Model: ldsjmdy/Tulu3-Block-FT in bfloat16; torch 2.11, transformers 5.20; NVIDIA L4. All differences are computed on the same questions, with 95% bootstrap ranges.